

TEZPUR UNIVERSITY

Autumn Semester End Examination, December, 2016

CO509: Data Mining and Data Warehousing

Full Marks: 60

Time: 2 hours

1. Fill up the blanks: 1×5=5
- A decision tree is called _____ if training error is low but testing error is high.
 - The _____ clustering algorithm finds nested clusters.
 - The measures in a data cube are always _____.
 - In data warehousing the _____ server precomputes cuboids containing summarized data.
 - The apriori algorithm is also called _____ algorithm.
2. Provide brief answers: 2×5=10
- What are the different attribute selection measures for constructing a decision tree?
 - State the different applications of a data warehouse.
 - Which clustering algorithms can be used to find clusters with arbitrary size and shape?
 - List some characteristics of partitional clustering algorithms.
 - What is a FP-tree? State its importance.
3. (a) What is a decision tree? 2
- (b) How are decision trees used for classification? 2
- (c) Present the generic algorithm for decision tree construction. 4
- (d) How do you evaluate the performance of a classifier? 4
4. (a) Define support and confidence in connection with association rule mining. 2
- (b) Describe the two steps of the apriori algorithm for finding frequent itemsets. 4
- (c) Mention two disadvantages of the apriori algorithm. 2
5. (a) Describe the concept of a data cube. 3
- (b) How is an n -dimensional cuboid displayed? 2
- (c) What do you understand by concept hierarchy? Give examples. 2
- (d) Briefly describe the typical OLAP operations in a data cube. 5
- (e) What is meant by materialization of a data cube? What are the choices for it? 3
6. Consider the following base table of OLAP data. Represent it using bitmap indexing. 3

RID	Item	City
1	Computer	Kolkata
2	Printer	Kolkata
3	Scanner	Kolkata
4	Computer	Mumbai
5	PrinterScanner	Mumbai
6	Scanner	Mumbai

7. Given the following transaction dataset:

P.T.O.

TID	Items Purchased
1	a, b, c, d
2	a, e, f
3	a, c, f
4	a, c, d, f
5	a, c

- (a) Find all frequent itemsets using an algorithm of your choice. Show the computation steps.
Take minimum support (minsup) = 40%. 5
- (b) Find two strong association rules from any one of the frequent 3-itemsets you have found.
Take minimum confidence (minconf) = 70%. 2

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